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The Russia–China Energy and Financial Axis

A Public-Source Assessment of Enforcement Resilience, Node Criticality, and Sanctions Friction Across a 60-Node Infrastructure Complex

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STRATEGIC ANALYSIS · SIMULATED · NOT CLASSIFIED · FOR ACADEMIC AND POLICY RESEARCH PURPOSES ONLY

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SCOPE AND ANALYTIC POSTURE

ANALYTIC POSTURE

This paper assesses the Russia–China strategic economic partnership as a sanctions-resilient and enforcement-adaptive architecture. Its principal analytic contribution is a two-axis node-criticality framework combining irreplaceability and throughput to distinguish high-value but substitutable nodes from nodes whose restriction or denial could generate more durable enforcement friction. The paper does not provide operational direction. It evaluates the public record, identifies areas of enforcement vulnerability and structural constraint, and assesses the policy relevance of different node classes under stated assumptions.

Part VI of this paper presents a scenario-conditioned sequencing model. That model is an analytic construct intended to make interdependency logic explicit and testable. It is not an implementation plan, a designation recommendation, or an assessment of operational readiness. The distinction between analytical completeness and operational maturity is addressed explicitly in Part VIII.

Analytic Purpose

This assessment addresses three questions. First, what does the public record establish about the architecture and resilience characteristics of the Russia–China energy and financial partnership as of Q1 2026? Second, which node classes within that architecture present the most analytically significant opportunities for enforcement friction, and on what evidentiary basis? Third, under what conditions and with what coalition dependencies would pressure combinations be most likely to generate durable effects?

The paper does not assess whether any specific enforcement action should be taken. That determination rests with relevant authorities. The paper assesses where the public record is sufficient to support provisional analytic judgments and where intelligence gaps, legal-operational distinctions, and coalition-dependence constraints limit the strength of those judgments.

Analytic Standards

All claims are sourced at point of use. Source tier citations follow the format [TIER 1 — source, date] or [TIER 2 — source, date]. Figures without Tier 1 or Tier 2 sourcing do not appear in this document. Probability and confidence are assessed on separate axes per ODNI Analytic Standards. Probability describes the likelihood a judgment is correct. Confidence describes the quality and completeness of supporting evidence.

CONFIDENCE LEVEL	EVIDENCE BASIS	PAIRED PROBABILITY
HIGH CONFIDENCE	Multiple independent Tier 1 sources, consistent findings, minimal inference	Almost certainly · Very likely
MODERATE-HIGH CONFIDENCE	Strong Tier 1/2 sourcing, limited acknowledged inference	Very likely · Likely
MODERATE CONFIDENCE	Credible Tier 1/2 sourcing,	Likely · Probably · Roughly even odds

	meaningful inference or unresolved variable	
LOW-MODERATE CONFIDENCE	Thin but structurally sound inference from indirect evidence	Unlikely · Roughly even odds
LOW CONFIDENCE	Primarily analytic inference; alternatives remain plausible	Unlikely · Very unlikely
ANALYTICALLY INFERRED	Derived by logical extension from documented premises; not directly observed	Any probability label — add to confidence tag
SCENARIO-CONDITIONED	Valid only under stated scenario premises; not a prediction	Any probability label — add to confidence tag

KEY JUDGMENTS

1

HIGH CONFIDENCE · DOCUMENTED

Official reporting and open-source commercial analytics support a high-confidence judgment that the Russia–China strategic partnership has developed a partial sanctions-resilient architecture spanning energy, minerals, finance, and digital communications. The A7 Network — combining the A7A5 stablecoin, the mBridge/CIPS interbank corridor, and a distributed crypto OTC layer — appears structured in ways that reduce exposure to Western decoupling pressure across identified transaction categories. The public record does not establish full economic autarky or immunity from external enforcement pressure [TIER 1 — OFAC, 2024–2025; TIER 2 — TRM Labs, 2025].

2

HIGH CONFIDENCE · DOCUMENTED

A principal constraint on Western enforcement effectiveness is assessed with high confidence to be adaptation speed and third-country enforcement deficits rather than legal authority. The United States holds ample designation power under existing executive and statutory frameworks. Open-source reporting on the Garantex-to-ABCeX migration pattern following the March 2022 OFAC designation indicates that entity-level action without simultaneous infrastructure restriction is likely to produce 30-day migration cycles with no durable disruption to network throughput [TIER 1 — OFAC designation records; TIER 2 — Chainalysis, 2023].

3

MODERATE-HIGH CONFIDENCE · ANALYTICALLY INFERRED

Available public reporting suggests that timing and simultaneity of pressure measures may materially affect enforcement durability. For planning purposes, the strongest defensible judgment is that a coordinated approach compressing stablecoin liquidity restriction, hosting-provider interdiction, and secondary sanctions notices into a single operational window is very likely to generate more durable friction than sequential designation. This assessment does not establish a universally applicable timing threshold; the 14–30 day migration window is inferred from documented migration patterns, not from a validated simultaneous interdiction attempt.

4

HIGH CONFIDENCE · DOCUMENTED

Seven nodes in the assessed 60-node network are Chinese sovereign infrastructure or Chinese-licensed entities. Official reporting and established legal analysis support a high-confidence judgment that US unilateral designation of these nodes under current political conditions is assessed as likely counterproductive, generating retaliation risk across the bilateral economic relationship that is likely to exceed enforcement value. Available evidence is more consistent with multilateral coordination through EU/UK/Australia channels and private diplomatic engagement than with unilateral action for this node class [TIER 1 — EO 14024 §1(a)(iii); TIER 2 — CSIS, 2025].

5

MODERATE CONFIDENCE · ANALYTICALLY INFERRED

The GENIUS Act (Pub. L. 119-27, 139 Stat. 419) provides mandatory freeze authority over registered stablecoin issuers including Tether/USDT. The public record suggests the stablecoin compliance architecture established by the Act may represent a more

administratively accessible near-term pressure surface against the A7A5 wallet cluster than other available instruments, subject to evidentiary nexus requirements and implementing regulation finalization. Full administrative operability depends on OFAC blockchain analytics establishing evidentiary nexus to designated entities and on implementing regulations from FinCEN and OFAC, which had not been fully published as of March 2026. The BRICS Bridge and mBridge CBDC infrastructure operates outside registered stablecoin jurisdiction; no equivalent restriction mechanism exists for sovereign CBDC systems under current statutory authority.

6

MODERATE CONFIDENCE · ANALYTICALLY INFERRED

Physical infrastructure nodes assessed as Quadrant A — Skovorodino ESPO fork, Zabaykalsk grain terminal, Zvezda Shipyard, Sabetta LNG terminal — are assessed with moderate confidence as irreplaceable within a 90-day window. The strongest defensible judgment is that meaningful enforcement friction against these nodes within the sanctions domain requires maximum-pressure multilateral designation coordinated with EU/UK/Australia. Options outside the sanctions enforcement domain are outside the scope of this assessment.

PART I: OBSERVED ARCHITECTURE

1.1 Energy Flows and Physical Infrastructure

Official US government reporting and open-source energy data indicate that Russia and China have substantially expanded bilateral energy trade since 2022, with Russian crude oil and LNG increasingly redirected toward Chinese end-markets following European embargo. The IEA World Energy Outlook 2025 reports that the ESPO pipeline corridor through Skovorodino transited in excess of $60 billion annually in crude oil flows to Chinese refiners as of 2024–2025 [TIER 2 — IEA WEO 2025]. Sabetta LNG terminal shipments along the Sabetta-Beihai corridor represent approximately 20 million tons per year under partial EU and UK sanctions that have not been matched by US designation of shadow fleet tankers operating that route [TIER 2 — IEA WEO 2025].

These findings are consistent with a judgment that bilateral energy infrastructure has achieved a degree of routing redundancy sufficient to sustain throughput under current Western sanctions levels. The public record does not establish whether these flows are formally insulated from incremental pressure escalation or whether the observed resilience reflects residual Western market dependence and logistical constraints on Russia's part that could be exploited under a more coordinated enforcement posture.

1.2 Financial Settlement Channels

OFAC enforcement actions and open-source blockchain analytics document a layered financial settlement architecture operating in parallel with SWIFT-based correspondent banking. Three principal layers are identified in the public record.

The A7A5 stablecoin layer processed approximately $39 billion in 2025 flow volume per commercial blockchain analytics [TIER 2 — Chainalysis Crypto Crime Report 2025]. The majority of this flow is assessed with moderate confidence to represent conversion of sanctioned Russian energy proceeds into USDT for dual-use component procurement transiting Hong Kong and Shenzhen intermediaries. The precise composition of flows — between energy proceeds, defense procurement, and financial arbitrage — is a principal uncertainty; the public record does not permit a firm conclusion on flow decomposition.

The mBridge and CIPS interbank corridor provides central-bank-to-central-bank digital settlement outside the SWIFT system. Public reporting and BIS documentation indicate this layer handles a growing share of bilateral trade settlement. The SWIFT Institute estimated BRICS Bridge and mBridge settlement at approximately 40% or more of bilateral trade by 2025 [TIER 2 — SWIFT Institute, 2025]. US Treasury holds no direct interdiction mechanism against central-bank-to-central-bank digital settlement under current statutory authorities; the GENIUS Act provides authority over registered stablecoin issuers but does not reach sovereign CBDC systems.

1.3 Digital Infrastructure and Hosting Dependencies

OFAC designation records and open-source hosting analysis identify Aeza Group and Media Land as the primary hosting providers servicing ABCeX, Garantex successor infrastructure, and Aifory Pro [TIER 1 — OFAC, 2024–2025]. These entities operate within Russian domestic network infrastructure but maintain ASNs that peer with US-connected Tier 1 internet backbone providers, creating a potential IEEPA §1702 interdiction surface.

The significance of this dependency is that it constitutes a structural vulnerability in the crypto OTC layer that has not been exploited in prior enforcement actions. OFAC's 2022 Garantex designation and subsequent ABCeX actions proceeded against entities without simultaneously restricting hosting infrastructure. The documented migration pattern is consistent with the hypothesis that hosting layer continuity was the enabling condition for rapid wallet migration [TIER 2 — Chainalysis, 2023].

1.4 Third-Country Facilitation Exposure

The Kyrgyzstan (Grinex/Meer exchange infrastructure), UAE (correspondent institutions processing A7A5-to-fiat conversions), and Hong Kong (freight forwarder and electronics reseller networks) nodes share a structural feature: they operate in jurisdictions where US primary sanctions do not apply and secondary sanctions require either aggressive CAATSA implementation carrying diplomatic costs or FATF referral processes with 18–24 month timelines.

Available reporting is consistent with a judgment that the A7A5 architecture was designed around these jurisdictional gaps. The public record does not establish whether this design reflects formal coordination between Russian and Chinese state actors or emergent adaptation by private-sector intermediaries responding to enforcement signals.

PART II: NODE-CRITICALITY FRAMEWORK

2.1 Framework Design and Axis Selection

This assessment employs a two-axis node-criticality framework combining Irreplaceability and Throughput Volume. These axes were selected because prior enforcement analyses have used throughput alone as a prioritization criterion, producing designation priorities that conflate high-volume substitutable nodes — where entity-level action predictably produces migration rather than friction — with genuinely irreplaceable physical or institutional nodes where restriction carries durable structural consequences.

Irreplaceability is defined as the degree to which a sanctioned node can be replaced by a functionally equivalent successor within 90 days without material loss of throughput or capability. This definition is intentionally conservative: it captures near-term substitution capacity under realistic operating conditions, not theoretical long-run alternatives. Throughput Volume reflects annual flow value or strategic significance for non-financial nodes. For physical infrastructure, strategic significance includes criticality to energy routing, component supply, or digital backbone functions that are not directly monetized.

This framework is intended as a heuristic for distinguishing node classes that present different exposure profiles and different constraints on pressure durability. It does not establish that pressure against any individual node would produce decisive system-wide effects. Each quadrant classification is assessed against the available public record and carries an explicit confidence label. The classification does not substitute for interagency consequence analysis or classified collection.

2.2 Quadrant Assignments and Evidentiary Basis

	HIGH THROUGHPUT	LOW THROUGHPUT
IRREPLACEABLE	**QUADRANT A — HIGHEST-CRITICALITY NODES** Skovorodino ESPO Fork, Zabaykalsk Grain Terminal, Zvezda Shipyard, BitRiver (A7A5 compute), Amurskaya Substation, Sabetta LNG Terminal *Primary exposure type: coalition-dependent physical and institutional chokepoint exposure. The public record suggests durable pressure would depend less on unilateral US action than on coordinated allied alignment. This assessment does not establish whether such alignment is currently attainable.*	**QUADRANT B — SECONDARY CONSTRAINT NODES** Tomtor REE Deposit, Krasnokamensk Uranium Hub, Polar Express Cable Nodes, Arctic LNG 2 Plant *Primary exposure type: long-horizon capacity constraint exposure. The public record suggests these nodes present lower near-term throughput but meaningful future leverage as capacity ramps. This assessment does not establish the timeline at which they would become highest-criticality.*
SUBSTITUTABLE	**QUADRANT C — HOSTING-LAYER DEPENDENCY NODES** ABCeX, Garantex mirror wallets, Aifory Pro, Hong Kong OTC cluster (freight forwarder tier)	**QUADRANT D — MONITORING PRIORITY** Individual HK shell OTC desks (Woeroon, Abingo), Erenhot Rail Hub, PoS-2 Planned

	Primary exposure type: infrastructure-continuity and liquidity-layer exposure. Available evidence suggests hosting-layer continuity and stablecoin liquidity access are more consequential to network durability than entity-level naming alone, based on documented migration patterns. The public record does not establish the disruption duration achievable under a more coordinated restriction approach.	Corridor *Primary exposure type: contingent exposure requiring further attribution. Available reporting does not permit a firm conclusion on network significance until exposure chain to a Quadrant A node is established. Monitoring priority only under current evidence.*

2.3 What the Framework Does Not Establish

METHODOLOGICAL LIMITATION

The two-axis framework is a heuristic for distinguishing node classes by exposure profile and pressure-durability constraint. It does not establish that pressure against any specific Quadrant A node would produce decisive system-wide effects. Node irreplaceability assessments are based on open-source analysis of physical infrastructure characteristics and commercial analytics on financial migration patterns; they have not been validated through interagency consequence modeling or classified collection.

Throughput figures reflect open-source energy data and commercial blockchain analytics. The crypto-layer figures understate total A7 network throughput because mBridge and direct CIPS settlement volumes are not accessible through open commercial analytics. The framework should be treated as a provisional prioritization model pending more complete collection.

PART III: LEGAL AUTHORITIES AND ADMINISTRATIVE OPERABILITY

3.1 Legal-Operational Distinction

Statutory authority and administrative operability are distinct. The United States holds ample designation power under existing executive and statutory frameworks against identified nodes. A principal constraint on converting that authority into durable enforcement friction is not textual but practical: evidentiary sufficiency, coalition alignment, timing and sequencing, and anticipatory adaptation by targets. This section maps legal authorities to node classes while explicitly distinguishing what the statutes authorize from what the public record supports regarding implementation feasibility.

AUTHORITY	PRIMARY APPLICATION	NEXUS REQUIREMENT	NETWORK NODES AND OPERABILITY NOTES
EO 14024 §1(a)(i)	Russian financial, tech, and energy sectors	Material support finding to designated sector or entity	ABCeX, Garantex successors, BitRiver, Aifory Pro, VTB Shanghai branch, ESPO infrastructure. The public legal record appears to support designation authority for these nodes; durable effect would require multilateral alignment for physical nodes and carries extraterritorial complexity for Chinese-licensed entities.
EO 14024 §1(a)(iii)	Technology transfer to Russia's military-industrial complex	Knowing facilitation of transactions in blocked sectors	Hong Kong OTC cluster, Aifory Pro USDT card network, Shenzhen component suppliers. The public legal record appears to support a knowing-facilitation nexus for these nodes; extraterritorial operability cannot be established from open sources alone and would require HK/PRC coordination for durable effect.
EO 13382	WMD proliferation networks	Nexus to nuclear, missile, or delivery system programs	Krasnokamensk uranium hub, Tianwan/Xudapu NPP fuel supply chain. The public legal record appears to support a proliferation-nexus basis; implementation feasibility cannot be established from open sources alone and requires an evidentiary chain to weapons programs that the public record does not fully characterize.
CAATSA §226	Secondary sanctions on foreign financial institutions (FFIs)	FFI must knowingly facilitate a significant financial transaction for a designated person	Keremet Bank (KG), Capital Bank (KG), UAE correspondent banks. The public legal record appears to support secondary designation authority where knowing facilitation of significant transactions is established; diplomatic costs of aggressive secondary designation represent a significant variable that the public record does not resolve.

GENIUS Act Pub. L. 119-27 §4(b)	Mandatory freeze authority over registered stablecoin issuers	Statute compels registered issuers including Tether/USDT to freeze OFAC-flagged wallets	A7A5 wallet cluster (~$39B 2025 flows). Authority enacted [TIER 1 — 139 Stat. 419]; FinCEN/OFAC implementing regulations pending as of March 2026. Evidentiary nexus to designated entities required.
IEEPA §1702	Emergency economic powers	Property interest of foreign country or national during declared national emergency	Aeza Group, Media Land hosting ASNs. Standing emergency active under EO 13694/14024 [TIER 1 — EO 13694; 90 FR 24723]. The public record suggests a legal basis may exist for BGP route withdrawal under IEEPA §1702, with prior analogous actions providing some precedent; implementation feasibility and interagency coordination requirements cannot be fully assessed from open sources alone.

3.2 VTB Shanghai: Implementation-Friction Case Study

ATTRIBUTION	OFAC designated VTB Bank under EO 14024 in February 2022 [TIER 1 — OFAC designation record]. The Shanghai branch operates under Chinese regulatory jurisdiction (National Financial Regulatory Administration license). No US secondary designation of a Chinese-licensed subsidiary of a Russian-designated bank has been executed to date.
JUDGMENT	The public record supports a high-confidence judgment that the VTB Shanghai branch presents an implementation-friction problem rather than a pure legal-authority problem. Because the branch operates within PRC regulatory jurisdiction, unilateral US action is likely to be less operationally meaningful than a coordinated multilateral approach and carries significant retaliation risk across the full bilateral economic relationship. Available evidence is more consistent with coordinated EU/UK/Australia engagement and private diplomatic dialogue through the US-China economic working group track than with unilateral action for this node.
LIMITS	The public record does not establish whether a coordinated multilateral designation track for VTB Shanghai has been pursued, at what level, or with what response from Beijing. The available reporting does not permit a firm conclusion on PRC tolerance thresholds for extraterritorial action against Chinese-licensed entities.

PART IV: ENFORCEMENT FRICTION ASSESSMENT

4.1 Adaptation and Substitution Dynamics

ATTRIBUTION	OFAC designated Garantex under EO 14024 in March 2022. Chainalysis documented migration of wallet infrastructure to ABCeX and successor entities within weeks. The Rapira exchange emerged as a secondary node within the same 30-day window [TIER 1 — OFAC designation record; TIER 2 — Chainalysis, 2023].
JUDGMENT	These findings support a high-confidence judgment that entity-level designation without simultaneous infrastructure restriction very likely produces a 30-day migration cycle with no durable disruption to network throughput. The strongest defensible inference from this pattern is that hosting-layer continuity was the enabling condition for rapid migration: because Aeza Group servers remained operational, wallet infrastructure could migrate without loss of network reachability. Available public reporting suggests that timing and simultaneity of pressure measures may materially affect enforcement durability.
LIMITS	The public record does not establish whether any prior enforcement action deliberately attempted simultaneous infrastructure restriction and was defeated, or whether sequential approaches reflect institutional constraints. This assessment does not establish a universally applicable migration timeline; the 14–30 day window is inferred from documented patterns, not from a validated simultaneous restriction attempt.

4.2 Coalition Dependence and Diplomatic Constraints

Official reporting and open-source analysis indicate that the enforcement impact of US unilateral action varies significantly by node class. For financial and digital infrastructure nodes in US-accessible jurisdictions, unilateral action is generally sufficient to initiate disruption but insufficient to prevent migration without coalition support. For physical infrastructure nodes and Chinese-licensed entities, the analytical record supports a judgment that US unilateral action is likely to produce less durable friction than coordinated multilateral action.

EU and UK sanctions architecture provides meaningful complementary coverage for Russian energy and financial infrastructure nodes, but enforcement variance across EU member states remains a documented constraint [TIER 2 — CSIS, 2024–2025]. Australia's designation authority aligns with US and UK frameworks for most node classes but carries limited independent enforcement leverage against Chinese-jurisdiction nodes. Japan and South Korea represent significant gaps in the multilateral coverage picture for energy transit and component supply chain nodes. The public record does not establish the conditions under which Japan or South Korea would align on coordinated designation of Russian physical infrastructure or Chinese-facilitated evasion nodes.

4.3 Extraterritorial Sensitivities

Three node classes present distinct extraterritorial sensitivity profiles that constrain enforcement instrument selection.

Kyrgyzstan nodes (Grinex/Meer exchange infrastructure): Available evidence suggests institutional-level secondary pressure under CAATSA §226 is more consistent with the jurisdictional constraints and diplomatic sensitivities involved than sovereign-level action, which appears likely to disrupt CSTO/SCO relationships without proportionate enforcement benefit. The public record does not establish whether institutional-level designation alone would be sufficient to materially reduce A7A5 issuance capacity or whether successor infrastructure has been pre-positioned in alternative jurisdictions.

UAE correspondent institutions: The UAE was grey-listed by FATF in March 2022 and removed in February 2024 following documented AML reforms [TIER 1 — FATF Public Statement, February 2024]. This track record is consistent with a judgment that bilateral diplomatic pressure is likely to produce meaningful institutional compliance before FATF re-referral becomes necessary. The UAE's demonstrated responsiveness to FATF processes is the evidentiary basis for this assessment.

Hong Kong OTC cluster: Hong Kong nodes present the highest extraterritorial complexity because enforcement action requires either PRC regulatory cooperation, which the public record does not support as likely under current conditions, or accepting that enforcement will be incomplete and focused on Western-accessible transaction layers only.

4.4 Infrastructure Visibility Gaps

IMPLEMENTATION FRICTION — THROUGHPUT VISIBILITY

The most significant enforcement visibility gap is the mBridge and CIPS settlement layer. Central-bank-to-central-bank digital settlement volumes are not accessible through open commercial blockchain analytics. The $39 billion A7A5 crypto-layer figure represents only the portion of bilateral financial flows accessible to open-source analysis. Total bilateral settlement volume including CIPS and direct correspondent channels is substantially higher and not visible through current open-source means.

This visibility gap has a direct implication for enforcement design: pressure measures targeting the A7A5 crypto layer would address a meaningful but not dominant share of total bilateral financial flows. The mBridge/CBDC layer, which is not reachable through GENIUS Act stablecoin authority, would remain operationally intact. The most policy-relevant implication is that stablecoin-centric enforcement should be understood as a friction measure against one layer of the architecture, not a comprehensive settlement disruption.

4.5 Risks of Premature Signaling

Available reporting on prior enforcement actions against crypto exchange infrastructure indicates that public designation announcements before infrastructure-level restrictions are in place provide operationally significant warning to wallet holders. The Garantex pattern is consistent with this dynamic: the March 2022 OFAC announcement preceded infrastructure action, enabling migration before the designation took full effect [TIER 2 — Chainalysis, 2023].

For planning purposes, the strongest defensible judgment is that enforcement sequencing that permits advance warning is likely to reduce disruption durability regardless of the legal completeness of the designation. The public record does not establish whether current OFAC and

interagency procedures include provisions for simultaneous infrastructure coordination or what the internal process requirements are for such coordination.

PART V: COUNTRY AND ACTOR SENSITIVITY ANALYSIS

5.1 Russian Node Classes

Russian physical infrastructure nodes in Quadrant A — ESPO fork, Zabaykalsk terminal, Zvezda Shipyard, Sabetta LNG terminal — are assessed as highest-criticality on both axes. Zvezda Shipyard presents a distinct supply-chain exposure: it is dependent on Chinese inputs for Arc7-class LNG carrier construction, assessed at 3–5 vessels per year [TIER 2 — CSIS, 2024]. Available evidence suggests this dependency lacks near-term Russian substitutes; the diplomatic cost of pressure on Chinese industrial suppliers is a significant variable the public record does not resolve.

BitRiver (Bratsk) presents a dual-use compute profile: it provides the primary proof-of-work infrastructure for A7A5 stablecoin issuance and is collocated with Bratsk Hydroelectric Plant, which provides cheap power with no viable equivalent site in Russia at comparable scale [TIER 2 — CSIS, 2024]. This combination of energy cost advantage and compute concentration supports a high-confidence assessment of near-term irreplaceability.

5.2 PRC-Jurisdiction Nodes

Seven nodes in the assessed 60-node network are Chinese sovereign infrastructure or Chinese-licensed entities: Ganzhou REE hub, Daqing/Dushanzi refineries, Tianwan/Xudapu NPPs, and VTB Shanghai branch. US designation authority formally applies to some of these under EO 14024 secondary sanctions provisions. The public record supports a high-confidence judgment that exercising that authority against Chinese sovereign industrial infrastructure under current conditions is assessed as likely counterproductive.

The Ganzhou REE hub warrants separate treatment. It constitutes a global monopoly position in rare earth element processing with no near-term allied substitute [TIER 2 — CSIS, 2025]. Available evidence suggests the structural dependency on this node is more amenable to long-term allied stockpiling and processing capacity development than to designation action, which the public record suggests would likely produce retaliation without delivering enforcement friction on the target node. The public record does not establish the timeline or cost at which allied processing alternatives could meaningfully reduce this structural dependency.

5.3 Hong Kong, UAE, and Central Asia Intermediary Exposure

The Hong Kong OTC cluster, UAE correspondent institutions, and Kyrgyz exchange infrastructure (Grinex/Meer) each present distinct sensitivity profiles that the available evidence suggests are not well-addressed by entity-level designation alone. These nodes are assessed as Quadrant C: substitutable within a 30-day window under entity-only pressure, but potentially subject to more durable friction under approaches that address hosting continuity, stablecoin liquidity, and secondary institutional exposure simultaneously.

The key analytical finding for this node class is that sequential designation — the pattern documented in prior enforcement actions — is likely to produce adaptation rather than durable friction. For planning purposes, available evidence suggests that approach simultaneity is more consequential for friction durability in this node class than the legal completeness of individual designations. The public record does not establish the degree to which simultaneity alone would overcome the jurisdictional gaps characterizing this node class.

5.4 Private-Sector Infrastructure Dependencies

Tier 1 internet backbone providers (Lumen Technologies, Cogent Communications, NTT Communications, Zayo Group) maintain peering relationships with ASNs associated with Aeza Group and Media Land. IEEPA §1702 provides authority for a Presidential directive requiring BGP route withdrawal for these ASNs under the standing national emergency declared under EO 13694 [TIER 1 — EO 13694; 90 FR 24723]. The public record suggests a legal basis may exist for this approach, with prior analogous IEEPA actions providing some precedent; implementation feasibility and interagency coordination requirements cannot be fully assessed from open sources alone.

Tether Operations Limited represents the critical private-sector dependency in the stablecoin layer. Open-source reporting documents Tether's compliance with prior freeze orders exceeding $100 million in 2025–2026 [TIER 2 — Tether transparency disclosures, 2025–2026]. This compliance track record supports a moderate-high confidence assessment that a GENIUS Act §4(b) freeze order would produce meaningful liquidity restriction on the identified wallet cluster, subject to evidentiary nexus requirements. The public record does not establish whether Tether's compliance posture would extend to a wallet cluster of the scale assessed here, or whether implementing regulations would impose procedural requirements affecting the operational timeline.

PART VI: SCENARIOS FOR DEFENSIVE PLANNING

SCENARIO MODEL SCOPE NOTE

The following scenarios are analytic constructs intended to illustrate how interdependency logic and instrument selection interact under different enforcement conditions. They are not implementation plans, operational directives, or assessments of operational readiness. Each scenario is scenario-conditioned: it describes a hypothetical enforcement posture and assesses likely second-order effects based on documented network behavior and migration patterns. The scenarios do not predict what will happen; they assess what the public record suggests about relative friction durability under different approaches.

Scenario A: Financial-Rail Pressure Intensification

Scenario premise: US authorities pursue intensified entity-level designation of ABCeX, Aifory Pro, and Grinex without simultaneous hosting or stablecoin restriction. This is the pattern most consistent with prior enforcement actions.

Available reporting on the Garantex-to-ABCeX migration is consistent with a judgment that this approach is likely to produce a 30-day migration cycle to successor infrastructure. The A7A5 wallet cluster would very likely migrate to non-USDT instruments or successor exchange infrastructure before designation takes full operational effect. The principal uncertainty is whether any successor infrastructure has been pre-positioned; the public record does not establish this. This scenario is assessed as likely to produce short-term disruption followed by rapid adaptation, consistent with the documented pattern.

Scenario B: Coordinated Hosting and Stablecoin Restriction

Scenario premise: IEEPA §1702 BGP withdrawal orders and GENIUS Act §4(b) mandatory freeze orders are initiated in a coordinated operational window, without advance public announcement of entity designation.

For planning purposes, the strongest defensible judgment is that this approach is more likely to generate durable enforcement friction than Scenario A, because it addresses the hosting-layer continuity and stablecoin liquidity conditions that enabled rapid migration in documented prior cases. The principal uncertainty concerns the 14–30 day migration window to Tor-accessible mirrors and .onion addresses, which represents the anticipated adaptation response. Available reporting suggests this migration is technologically feasible but operationally degraded relative to clearnet exchange access [TIER 2 — Chainalysis, 2023]. The public record does not establish the duration of the window during which liquidity restriction would remain effective, nor whether that window would be sufficient to achieve durable disruption.

This scenario's feasibility depends on GENIUS Act implementing regulation finalization, OFAC blockchain analytics establishing evidentiary nexus to designated entities, and interagency coordination between Treasury, Commerce, and the National Security Council for simultaneous instrument deployment.

Scenario C: Multilateral Coordination Shortfall

Scenario premise: US initiates maximum-pressure multilateral designation of Quadrant A physical nodes but EU/UK/Australia coordination is delayed or partial.

Official reporting on prior Russia sanctions implementation indicates that enforcement variance across EU member states and between US and EU designation timelines has historically reduced the cumulative pressure of multilateral campaigns [TIER 2 — CSIS, 2024–2025]. Under this scenario, Russian physical infrastructure nodes would likely face US designation but continued access to EU-adjacent financial and logistics infrastructure during the coordination gap. The most policy-relevant implication is that coordination timeline appears to be a principal constraint on physical node pressure campaigns, not legal authority. This assessment does not establish what coordination timelines are achievable under current allied consultation structures.

Scenario D: Rapid Migration to Alternative Settlement Paths

Scenario premise: Enforcement pressure on the A7A5 crypto layer produces accelerated migration to mBridge/CIPS settlement and direct PRC-Russia correspondent banking, reducing reliance on USDT-denominated flows.

This scenario represents the most significant medium-term adaptation risk. The mBridge and CIPS infrastructure is outside current stablecoin enforcement jurisdiction and does not depend on Western correspondent banking relationships or Tier 1 internet backbone peering. If A7A5 pressure accelerates migration to this layer rather than disrupting bilateral settlement, enforcement effectiveness could be reduced below baseline. The public record does not establish the current share of bilateral flows already transiting mBridge/CIPS versus the A7A5 layer, which is the principal uncertainty determining the severity of this risk.

PART VII: CONSOLIDATED NODE ASSESSMENT

NODE	QUADRANT	IRR.	THROUGHPUT	SUBSTITUTED?	ASSESSMENT AND PRIMARY EXPOSURE TYPE
Skovorodino ESPO Fork	A	High	$60B+/yr [TIER 2 — IEA, 2025]	No	Assessed as highest-criticality on both axes. Node appears most sensitive to maximum-pressure multilateral designation; available evidence suggests durable friction would require EU/UK/AUS coordination. Coalition-dependent exposure.
Zabaykalsk Grain Terminal	A	High	>5M tons/yr [TIER 2 — FAO, 2024]	No	Available evidence suggests this node is more exposed to buyer-side secondary pressure than direct designation; targeting logistics buyers transiting the terminal appears more consistent with WTO constraints than direct terminal action. Coalition-dependent exposure.
Zvezda Shipyard	A	Extreme	3–5 Arc7/yr [TIER 2 — CSIS, 2024]	No	Node appears most sensitive to Chinese input supply chain restriction; available evidence suggests Arc7 production depends on Chinese steel and navigation system inputs with no near-term Russian substitute. Supply-chain exposure; diplomatic cost of Chinese supplier action is a significant variable the public record does not resolve.
BitRiver (Bratsk)	A/B	High	A7A5 compute infrastructure	No	Node appears sensitive to dual-use compute restriction under EO 14024 §1(a)(i). Collocated power dependency with Bratsk Dam supports a high-confidence near-term irreplaceability assessment. Physical infrastructure exposure; no viable alternative site at scale identified in the public record.
Sabetta LNG Terminal	A	High	~20M tons/yr [TIER 2 — IEA, 2025]	No	Node appears sensitive to shadow fleet tanker restriction; available evidence suggests continued operation depends on tanker access not currently covered by US designation. Already under partial EU/UK sanctions. Coalition-dependent exposure; gap in US designation coverage is documented.
ABCeX (Federation Tower)	C	Low-Med	~$11B/yr [TIER 2 — TRM Labs, 2025]	Yes (ex-Garantex)	Node appears more sensitive to hosting-layer restriction and stablecoin liquidity denial than to entity designation alone. Available evidence suggests entity-only action is likely to repeat the documented Garantex migration pattern. Hosting-layer and stablecoin-liquidity exposure.
Garantex mirror wallets	C	Low	~$10B/yr [TIER 2 — Chainalysis, 2023]	Yes	Node appears sensitive to graph-based wallet tracing and stablecoin liquidity restriction on identified cluster. The public record suggests successor address pre-

					positioning is a documented adaptation pattern. Stablecoin-liquidity exposure; monitoring priority for successor address identification.
Grinex/Meer (Bishkek)	C	Med	A7A5 issuance infrastructure	Partial	Node appears sensitive to institutional-level secondary pressure under CAATSA §226; available evidence suggests sovereign-level action would be counterproductive relative to institutional targeting. Jurisdiction-gap exposure; third-country facilitation constraint.
Hong Kong OTC Cluster	C	Med	Dual-use hardware procurement	Partial	Node class appears more sensitive to simultaneous restriction across freight forwarder and electronics reseller tiers than to sequential action; available evidence suggests sequential designation permits adaptation before restrictions take effect. Graph-mapping required to characterize full network exposure before further assessment. Sequencing-dependent exposure.
VTB Shanghai Branch	A	Extreme	Dark clearing operations	No	Very high diplomatic-friction exposure with low apparent unilateral leverage. The public record supports a high-confidence judgment that this node's significance lies in coalition and jurisdictional constraints rather than in the absence of legal authority. Available evidence is more consistent with coordinated multilateral engagement than with unilateral action. The available reporting does not permit a firm conclusion regarding Beijing's tolerance threshold or allied willingness to align on this node class.
Ganzhou REE Hub	B	Extreme	Global monopoly position	No	Outside assessed US unilateral enforcement appetite under current conditions. Most policy-relevant implication from available evidence: allied stockpiling and processing capacity development addresses the structural dependency more durably than designation. Low unilateral leverage; structural exposure only.
Polar Express Cable Nodes	B	High	Digital backbone	No	No current US designation authority over Russian domestic cable infrastructure identified in the public record. Arctic-waters nodes may present a different exposure profile; the public record does not characterize this with precision. Monitoring priority; low unilateral leverage under current authorities.
BRICS Bridge/mBridge CBDC	A/B	Extreme	~40%+ bilateral settlement [TIER 2 — SWIFT Institute, 2025]	No	No short-term unilateral restriction mechanism identified in the public record under current statutory authorities. Available evidence suggests BIS multilateral engagement and long-lead settlement infrastructure development address the structural gap more directly

					than available near-term instruments. Structural exposure only; low-unilateral-leverage assessment.

PART VIII: INTELLIGENCE GAPS AND LIMITATIONS OF THE PUBLIC RECORD

8.1 Attribution and Evidentiary Limits

Wallet address attribution for the A7A5 cluster is based on open-source graph analysis from commercial blockchain analytics firms. The evidentiary standard that OFAC would apply to a mandatory Tether freeze order — and whether the current open-source attribution meets that standard — is a principal uncertainty. This assessment does not evaluate OFAC's internal evidentiary threshold. The available open-source attribution is sufficient to support a provisional analytical model, not a conclusion about designation readiness.

Financial flow figures for the A7A5 network represent USDT and crypto-layer flows only. mBridge and CIPS settlement volumes are not accessible through open commercial analytics. Crypto-layer figures therefore understate total A7 network throughput by an unknown margin. The public record does not permit a firm conclusion on the share of total bilateral financial flows transiting each settlement layer.

8.2 Migration Timeline Uncertainty

The 14–30 day migration window for exchange infrastructure to Tor-accessible mirrors is inferred from documented migration patterns following the Garantex designation [TIER 2 — Chainalysis, 2023]. This assessment has not been validated through an observed simultaneous infrastructure restriction attempt. The window could be shorter if successor infrastructure was pre-positioned, or longer if the simultaneous restriction of hosting infrastructure and stablecoin liquidity degraded migration capacity. The public record does not establish the sensitivity of migration timelines to hosting availability.

The simultaneity hypothesis — that compressing stablecoin restriction and hosting interdiction into a single operational window would prevent the documented migration pattern — is analytically inferred, not empirically confirmed. Reasonable alternative explanations exist, including that migration could proceed through alternative hosting providers not covered by the BGP withdrawal order, or that migration has already been partially completed in anticipation of future enforcement action.

8.3 Legal-Operational Gap

The GENIUS Act (Pub. L. 119-27, enacted 2025) provides mandatory freeze authority over registered stablecoin issuers. FinCEN and OFAC implementing regulations had not been fully published as of March 2026. Operational implementation of the mandatory freeze mechanism may depend on implementing regulations that are pending. This assessment treats GENIUS Act authority as enacted and analytically applicable to the stablecoin layer but notes that administrative operability under implementing regulations is a principal uncertainty that the public record does not resolve.

Statutory authority does not imply immediate enforceability. The gap between the legal authority mapped in Part III and operational enforcement friction involves evidentiary requirements, interagency coordination, coalition alignment, and target adaptation that the public record does not fully characterize. This assessment does not evaluate that gap as a finding about what enforcement actions should be taken.

8.4 PRC Internal Decision-Making

The degree to which observed Russia–China economic integration reflects deliberate state-directed strategy versus market-driven adaptation by Chinese private actors is a principal uncertainty that the public record does not resolve. Official US government reporting characterizes the partnership as strategically significant, but does not establish the precise role of PRC state direction in A7A5 architecture development, Hong Kong OTC cluster facilitation, or dual-use component supply chain organization.

This uncertainty is analytically significant because it affects the predicted response to enforcement pressure. If the architecture reflects deliberate PRC state direction, enforcement pressure on specific nodes may produce state-directed adaptation at speed and scale beyond what private-sector migration patterns suggest. The public record does not permit a firm conclusion on this question.

8.5 Analytical Boundary Statement

DOCUMENTED CONSTRAINT — ANALYTICAL SCOPE

The current public record is sufficient to support a provisional node-prioritization model and scenario-conditioned friction assessments. It is not sufficient to support a complete implementation picture. Intelligence and evidentiary gaps remain significant, particularly regarding mBridge/CIPS throughput, wallet attribution sufficiency for GENIUS Act operability, migration timeline sensitivity to simultaneous restriction, and PRC state-direction in network architecture.

The gap between this analysis and any assessment of operational readiness includes collection, evidentiary, coordination, and consequence-modeling questions that the public record does not resolve. The analytical scaffold identifies where enforcement friction appears most plausible and which node classes the available evidence suggests may be more or less exposed to different pressure approaches. It does not establish that any specific enforcement action would produce a predicted effect, nor does it resolve the institutional, diplomatic, and evidentiary requirements that would precede any implementation. Those are not analytical design questions that this paper can answer. A further constraint is that open-source network visibility may itself be a selection effect shaped by concealment, expendable-node exposure, or attribution distortion. Adversary architectures may be designed to over-represent commercially visible nodes while concealing higher-value state-directed infrastructure; some observed adaptation may reflect prior pre-positioning rather than reactive substitution; and open-source attribution chains may be vulnerable to deliberate shell layering or false-front facilitation that the public record cannot detect. This assessment does not establish whether any identified node class is part of a deliberate deception architecture or whether the visible network accurately represents the full enforcement-relevant surface. Node prioritization assessments in this paper should not be treated as counterintelligence findings.

PART IX: IMPLICATIONS FOR U.S. AND ALLIED POLICY

9.1 Principal Leverage Points

The available evidence is consistent with three areas where the public record suggests exposure to enforcement friction that may be more accessible than others. First, the Tether/USDT compliance architecture established by the GENIUS Act creates a stablecoin-layer restriction surface against the A7A5 wallet cluster; the public record suggests this surface may be more administratively accessible than other available approaches, subject to evidentiary nexus requirements and implementing regulation finalization. Second, the hosting-layer dependency of Quadrant C nodes on Aeza Group and Media Land infrastructure represents an exposure point addressable under IEEPA §1702 that the public record indicates has not been engaged in prior enforcement actions. Third, the Chinese input dependency of Zvezda Shipyard for Arc7-class construction represents a supply-chain exposure point; available evidence suggests this dependency lacks near-term Russian substitutes, though the diplomatic cost of pressure on Chinese industrial suppliers is a variable the public record does not resolve.

The most policy-relevant implication is that the most accessible leverage points are in the digital and financial layers, not the physical infrastructure layer. Physical Quadrant A nodes are highest-criticality but require coalition coordination that creates implementation friction. Digital and financial layer instruments are more unilaterally accessible but address a smaller share of total bilateral throughput and face documented migration dynamics.

9.2 Conditions Under Which Pressure Is More or Less Effective

Available evidence supports the following conditional assessments, each of which is scenario-conditioned and carries explicit confidence labels.

The public record is more consistent with durable enforcement friction under conditions where: instrument simultaneity reduces advance warning to wallet holders (MODERATE-HIGH CONFIDENCE, ANALYTICALLY INFERRED); multilateral designation is coordinated with EU/UK/Australia rather than executed unilaterally (HIGH CONFIDENCE, DOCUMENTED); secondary pressure targets institutional nodes rather than sovereigns in third-country jurisdictions (MODERATE CONFIDENCE, ANALYTICALLY INFERRED); and stablecoin restriction is combined with hosting-layer restriction rather than pursued independently (MODERATE CONFIDENCE, ANALYTICALLY INFERRED).

The public record is more consistent with rapid adaptation and reduced friction durability under conditions where: sequential designation provides advance warning enabling migration (HIGH CONFIDENCE, DOCUMENTED); enforcement action targets Quadrant C nodes without simultaneous infrastructure restriction (HIGH CONFIDENCE, DOCUMENTED); US acts unilaterally against Chinese-licensed entities without allied coordination (HIGH CONFIDENCE, DOCUMENTED based on retaliation risk); and enforcement pressure accelerates migration to mBridge/CIPS settlement layers outside stablecoin jurisdiction (MODERATE CONFIDENCE, ANALYTICALLY INFERRED).

9.3 Coalition Dependencies and Escalation Considerations

A principal coalition dependency identified in the public record is EU alignment on physical infrastructure designation. Available evidence suggests US designation of Quadrant A nodes in the absence of EU participation is unlikely to produce the throughput impact that makes those nodes highest-criticality: Russian energy infrastructure appears able to sustain operations if European logistics and financial services access remains partially available. The conditions under which the EU would accelerate designation alignment with the US on Russian physical infrastructure are outside the scope of this assessment.

Secondary pressure against Chinese-licensed entities or Chinese sovereign industrial infrastructure carries escalation risk across the full bilateral economic relationship. The public record does not support an assessment that enforcement value would exceed retaliation risk under current conditions; that balance is a policy judgment this paper does not reach. This does not mean such action is analytically excluded from consideration — it means the evidentiary basis for a contrary judgment is not present in the open-source record.

SOURCES AND REFERENCE BASIS

Tier 1 — US Government, UN Bodies, and Primary Legal Texts

- OFAC. Designation of Garantex under EO 14024. March 2022.
- OFAC. Designation of VTB Bank under EO 14024. February 2022.
- OFAC. Designation records for ABCeX and BitRiver. 2023–2025.
- GENIUS Act, Pub. L. 119-27, 139 Stat. 419 (2025). §4(b) mandatory freeze provisions.
- EO 14024. Blocking Property with Respect to Specified Harmful Foreign Activities of the Government of the Russian Federation. As amended by EO 14114.
- EO 13382. Blocking Property of Weapons of Mass Destruction Proliferators and Their Supporters.
- EO 13694. Blocking the Property of Certain Persons Engaging in Significant Malicious Cyber-Enabled Activities. As amended by EO 14306, 90 FR 24723.
- CAATSA §226. Countering America's Adversaries Through Sanctions Act.
- IEEPA §1702. International Emergency Economic Powers Act.
- FATF. Public Statement on UAE Removal from Grey List. February 2024.

Tier 2 — Policy Institutions, Commercial Analytics, and Specialized Publications

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- TRM Labs. Russia Sanctions Evasion Report. 2025. A7A5 flow volume and wallet cluster analysis.
- IEA. World Energy Outlook 2025. ESPO pipeline throughput and Sabetta LNG terminal volume figures.
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- SWIFT Institute. CBDC and mBridge Settlement Analysis. 2025. BRICS Bridge trade settlement share estimates.
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- WP-2026-MAXPRESS-01. Coordinated Maximum Pressure Campaigns: An Illustrated Execution Framework. Russia/shadow fleet track.
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- WP-2026-PERSIST-01. Litigation-Resilient Architecture for U.S. Competitive Statecraft. Six authority rails.
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